Econometrics

Dr. Mutua Samuel

Week 3 Notes

Contents

1 Multicollinearity

Perfect multicollinearity is the violation of the assumption of no multicollinearity.

MLR requires predictors to be **linearly independent** i.e no predictor can be expressed as a linear combination of others.

Multicollinearity means predictors in an MLR model have a close to exact linear relationship.

When multicollinearity exists the LS estimates for β_j^s exists but have a large variability.

1.1 Practical Consequences of Multicollinearity

- Although BLUE the OLS estimators have large variances and covariances making precise estimation difficult.

- Because of consequence 1, the CI tend to be much wider leading to acceptance of H_0 readily.
- The t ratio of one/more coefficients tends to be statistically insignificant.
- R^2 the overall measure of goodness of fit can be very high
- OLS estimates and standard errors can be sensitive to small changes in the data.

1.2 Detection of Multicollinearity

- High correlation coefficients: If correlation is above 0.8 then severe multicollinearity may be present
- High R^2 with low t-statistic values
- High variance inflation factors: The VIF above 10

1.3 Remedies for Multicollinearity

- Do nothing: live what you have
- Drop a redundant variable
- Transform the multicollinear variables
- Increase the sample size

2 Homoscedasticity

The variance of each error term e_i conditional on the explanatory values is some constant equal to σ^2

Symbolically $E[e_i^2] = \sigma^2$

2.1 Possible sources of heteroscedasticity

- Presence of outliers
- Skewness in the distribution of one or more regressors included in the model
- Incorrect data transformation
- Incorrect functional form

Problem of heteroscedasticity is likely to be more common in cross sectional data than in time series data.

2.2 Effects of heteroscedasticity

- The estimates are no longer BLUE and do not have minimum variance
- Since the OLS standard errors are based on the variances they are no longer valid in construction of confidence intervals.
- The CI are larger when there is heteroscedasticity

2.3 Remedies

- The Weighted Least Squares fixes the problem of heteroscedasticity.
- We can also transform the response variable but sometimes such transformation is not available

2.4 Weighted Least Squares

For the model

$$Y_i = \beta_0 + \beta_1 X_{i1} + ... + \beta_p X_{ip} + \epsilon_i$$

where ϵ_i are independent with $Var(\epsilon_i) = \sigma_i^2$ the WLS method finding estimates for $\beta's$ by minimizing

$$L(\beta_0, ..., \beta_p) = \sum_{i=1}^{n} \frac{(y_i - \beta_0 - ... - \beta_p x_{ip})^2}{\sigma_i^2}$$

In OLS $\sigma_i^2 = \sigma^2$ for all i, equivalent to minimizing

$$\sum_{i=1}^{n} (y_i - \beta_0 - ... - \beta_p x_{ip})^2$$

In WLS we focus on:

- More on minimizing errors of observations with smaller variances
- Less on minimizing errors of observations with larger variances

2.4.1 How to estimate unequal variance

The parameters in OLS: $\beta_0, \beta_1, ..., \beta_p, \sigma^2$

The parameters in WLS: $\beta_0, \beta_1, ..., \beta_p, \sigma_1^2, ..., \sigma_n^2$

Need prior knowledge about the variances σ_i^2. We will focus on the case when σ_i^2 are **inversely proportional** to some weights w_i

$$\sigma_i^2 = \frac{\sigma^2}{w_i}$$

where the weights $w_1, ..., w_n$ are known positive numbers and σ^2 is known. In this case the WLS is equivalent to minimize:

$$\sum_{i=1}^{n} w_i (y_i - \beta_0 - \beta_1 x_{i1} - ... - \beta_p x_{ip})^2$$

The WLS estimate of $\beta = (\beta_0, \beta_1, ..., \beta_p)^T$ that minimize the weighted sum of squares

$$\sum_{i=1}^{n} w_i (y_i - \beta_0 - \beta_1 x_{i1} - ... - \beta_p x_{ip})^2$$

is

$$\hat{\beta_{WLS}} = (X^T W X)^{-1} X^T W Y$$

where W is $n \times n$ matrix with $(w_1, ..., w_n)$ on the diagonal and 0 elsewhere.

2.4.2 Standard errors of WLS estimates

Under the model

$$Y_i = \beta_0 + \beta_1 x_{i1} + ... + \beta_p x_{ip} + \epsilon_i \;\; Var(\epsilon_i) = \frac{\sigma^2}{w_i}$$

the covariance matrix of $\hat{\beta_{WLS}}$ is:

$$Cov(\hat{\beta_{WLS}}) = \sigma^2 (X^T W X)^{-1}$$

The unknown variance parameter σ^2 is estimated by:

$$\hat{\sigma^2} = MSE = \frac{SSE}{n-p-1}$$

where

$$SSE = \sum_i w_i(y_i - \hat{y_i})^2$$

where the fitted values are:

$$\hat{y_i} = \beta_{0,\hat{WLS}} + \sum_{j=1}^{p} \hat{\beta}_{j,WLS} x_{ij}$$

The standard error of the WLS estimate $\hat{\beta}_{j,WLS}$ for βj is

$$\sqrt{\hat{\sigma^2} \times j^th\ \ diagonal\ \ element\ \ of\ \ matrix\ \ (X^TWX)^{-1}}$$

2.4.3 Tests and CIs for β_j for WLS

The t-statistic

$$\frac{\hat{\beta}_{j,WLS} - \beta_j^0}{se(\hat{\beta}_{j,WLS})} \sim t_{n-p-1}$$

under $H_0 : \beta_j = \beta_j^0$

and the t-CI

$$\hat{\beta}_{j,WLS} \pm t_{(n-p-1,\frac{\alpha}{2})} se(\hat{\beta}_{j,WLS})$$

2.5 WLS when σ_i is proportional to x_i

Suppose the variance of the i^{th} observation

$$\sigma_i^2 = Var(\epsilon_i) = \sigma^2 x_1^2$$

is known to be proportional to some value $x_i > 0$ where $\sigma^2 > 0$ is an unknown constant.

Since σ^2 is a constant, this is equivalent to use the weights

$$w_i = \frac{1}{x_i^2}$$

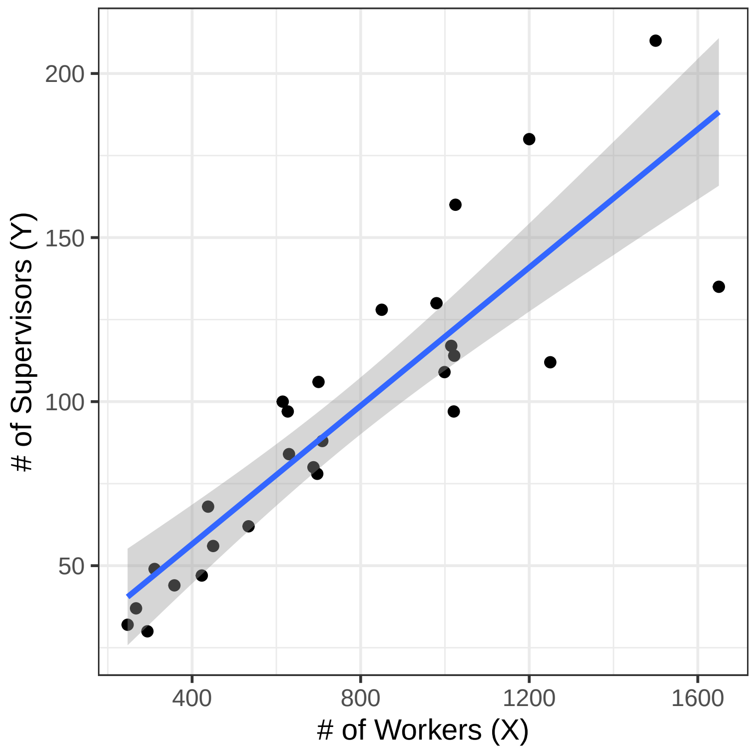
Thus we minimize:

$$L(\beta_0, \beta_1) = \sum_{i=1}^{n} \frac{1}{x_i^2}(y_i - \beta_0 - \beta_i x_i)^2$$

We consider the employee data

$X = Number\ \ of\ \ Supervised\ \ Workers\ Y = Number\ \ of\ \ Supervisors\ \ in\ \ 27\ \ Industrial\ \ Establishment$

```r
library(ggplot2)
supvis <- read.csv("supvis.csv")
ggplot(supvis, aes(x=X, y=Y))+geom_point()+geom_smooth(method='lm')+
labs(x="# of Workers (X)", y="# of Supervisors (Y)") + theme_bw()
```



As the variance of Y is proportional to X, we can use WLS the weight $w_i = \frac{1}{x_i^2}$

The `lm()` command can be used to fit WLS. One just needs to specify the weights in addition

```r
supvis <- read.csv("supvis.csv")
pander(summary(lm(Y ~ X, data=supvis, weights=1/X^2)))
```

	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept)	3.803	4.57	0.8323	0.4131
X	0.121	0.008999	13.45	6.044e-13

Table 2: Fitting linear model: Y ~ X

Observations	Residual Std. Error	R^2	Adjusted R^2
27	0.02266	0.8785	0.8737

2.5.1 Sum of Squares and Multiple R^2

- $$SST = \sum_i w_i(y_i - \bar{y_w})^2$$

  where $\bar{y_w} = \frac{\sum w_i y_i}{\sum w_i}$

- $$SSR = \sum w_i(\hat{y_i} - \bar{y_w})^2$$
- $$SSE = \sum w_i(y_i - \hat{y_i})^2$$
- SST = SSR + SSE remains valid
- df of SS same as OLS
- Multiple $R^2 = \frac{SSR}{SST}$. Cannot compare the multiple R^2 of a WLS model and OLS model since SSR and SST are calculated differently.
- $$MSE = \frac{SSE}{n-p-1} = \hat{\sigma^2}$$
- `Residual standard error` gives $\sqrt{MSE}$
- The estimate for $\sigma_i^2 = Var(\epsilon_i) = \frac{\sigma^2}{w_i}$

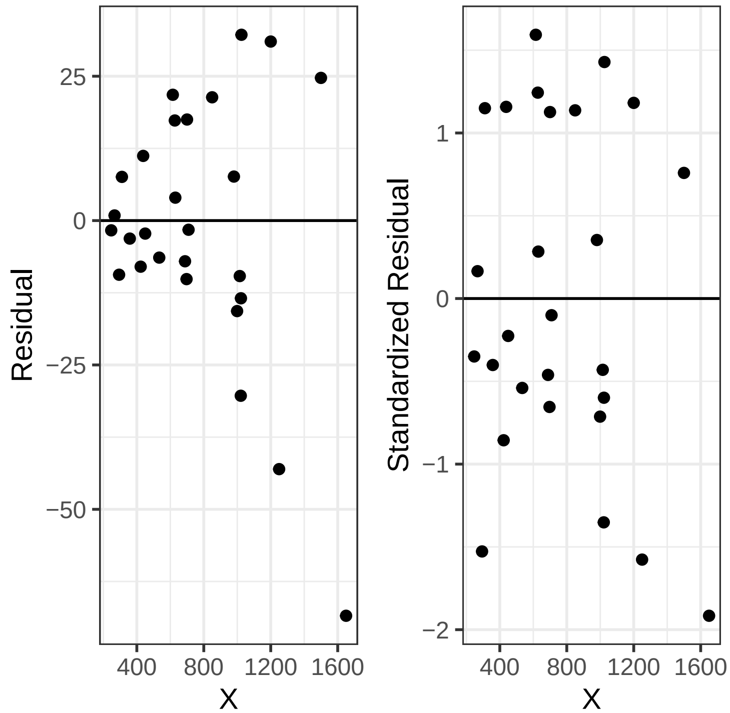
2.5.2 Residuals for WLS in R

- `model$res` gives the raw residuals $e_i = y_i - \hat{y_i}$ which are not adjusted by weight
- `hatvalues(model)` gives the leverage h_{ii} which is the i^{th} diagonal element of the hat matrix

$$H = X(X^TWX)^{-1}X^TW$$

- $Var(e_i) = \sigma_i^2(1 - h_{ii})$
- `rstandard(model)` gives internally studentized residuals
- `rstudent(model)` gives externally studentized residuals

```
lmwls = lm(Y ~ X, data=supvis, weights=1/X^2)
p1<- ggplot(supvis, aes(x=X, y=lmwls$res)) + geom_point() +
ylab("Residual") + geom_hline(yintercept=0) + theme_bw()
p2 <- ggplot(supvis, aes(x=X, y=rstandard(lmwls))) + geom_point() +
ylab("Standardized Residual") + geom_hline(yintercept=0)+theme_bw()
p1 + p2
```



The raw residuals are not weight-adjusted. The residual plot is still funnel-shaped

To see if the weights are chosen properly to fix the heteroscedastic problem, plot standardized or studentized residuals and see if the points scatter evenly around the zero line.

2.5.3 Confidence Interval/Predictions for WLS model

Note that weights must be provided for prediction or the intervals computed won't be correct.

```
predict(lmwls, data.frame(X=1200), weights=1/1200^2,
interval="confidence")
```

```
##        fit      lwr      upr
## 1 148.9917 134.2599 163.7235
```

```
predict(lmwls, data.frame(X=1200), weights=1/1200^2,
interval="prediction")
```

```
##        fit      lwr      upr
## 1 148.9917 91.07202 206.9113
```

At 95% confidence, industrial establishments with 1200 workers require 134.26 to 163.72 supervisors on average.

At 95% confidence, an industrial establishment that has 1200 workers is predicted to have 91.07 to 206.91 supervisors

3 Transformation of Variables

We transform variables including predictors and responses primarily for two reasons:

- To solve the non-linearity problem

- To solve the non-constant variability problem

3.1 Linear and Non Linear Models

Recall that linear models are linear in the parameters and not the predictors.

All the following are linear models:

- $Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \epsilon$
- $Y = \beta_0 + \beta_1 \sqrt{X} + \epsilon$

Whereas the following model is not linear since it is not linear in β_1

$$Y = \beta_0 + \exp(\beta_1 X) + \epsilon$$

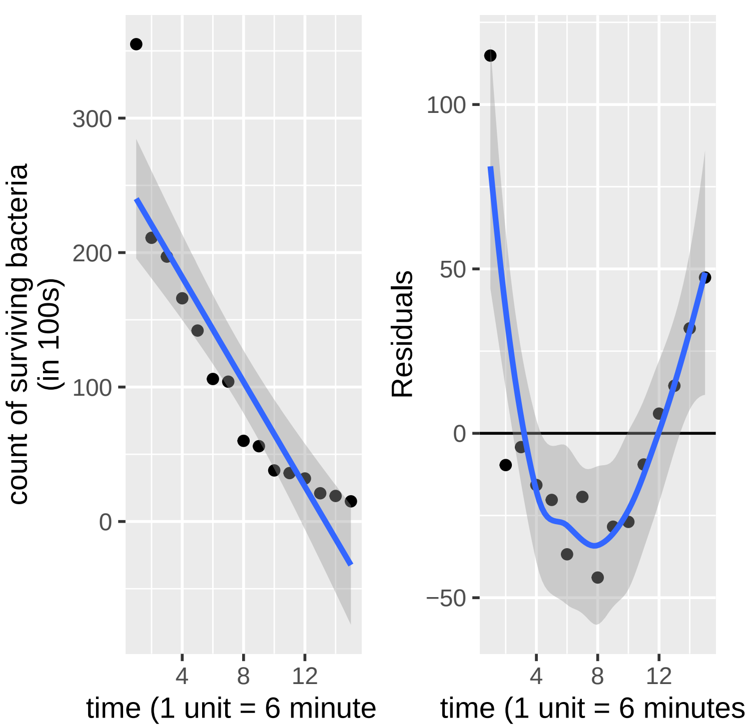
3.2 Transformation to achieve linearity

We use the bacteria data to demonstrate.

t time N_t the number of surviving bacteria following exposure after time t

We can fit a SLM and have

```
bact <- read.csv("bact.csv")
p1<- ggplot(bact, aes(x=t, y=N_t))+geom_point()+
geom_smooth(method='lm')+ xlab("time (1 unit = 6 minutes)")+
ylab("count of surviving bacteria\n(in 100s)")
lm1 = lm(N_t ~ t, data=bact)
p2<- ggplot(bact, aes(x=t, y=lm1$res))+geom_point() +
labs(x="time (1 unit = 6 minutes)", y="Residuals")+
geom_hline(yintercept=0) + geom_smooth()
p1 + p2
```



When not knowing what transformation to make, we would begin by looking at the scatterplot. In this case, the scatterplot is obviously non-linear.

According to theory, we expect a exponential decay in the count of bacteria in time:

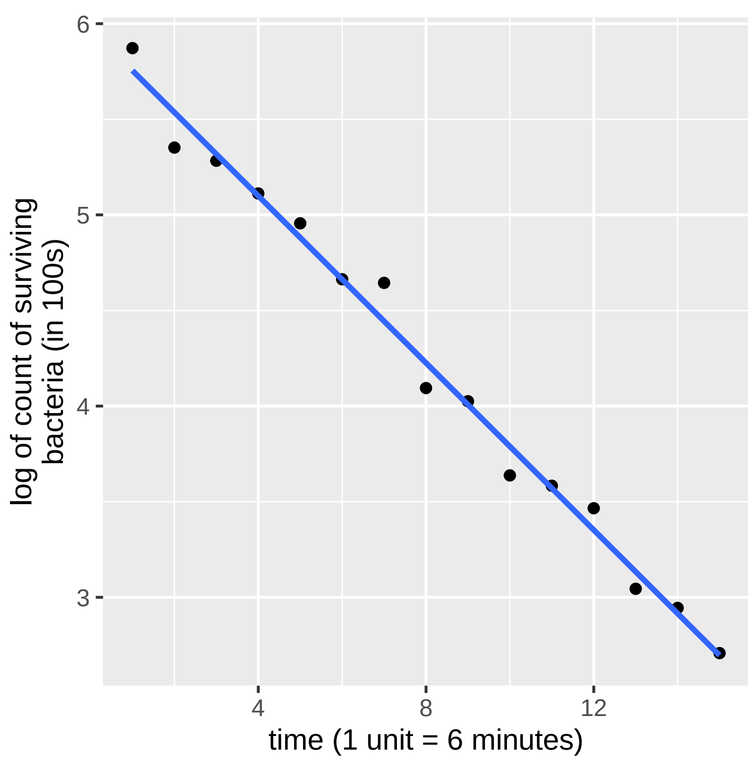
$$n_t = n_0 e^{\beta_1 t}$$

Taking logarithm on both sides we have:

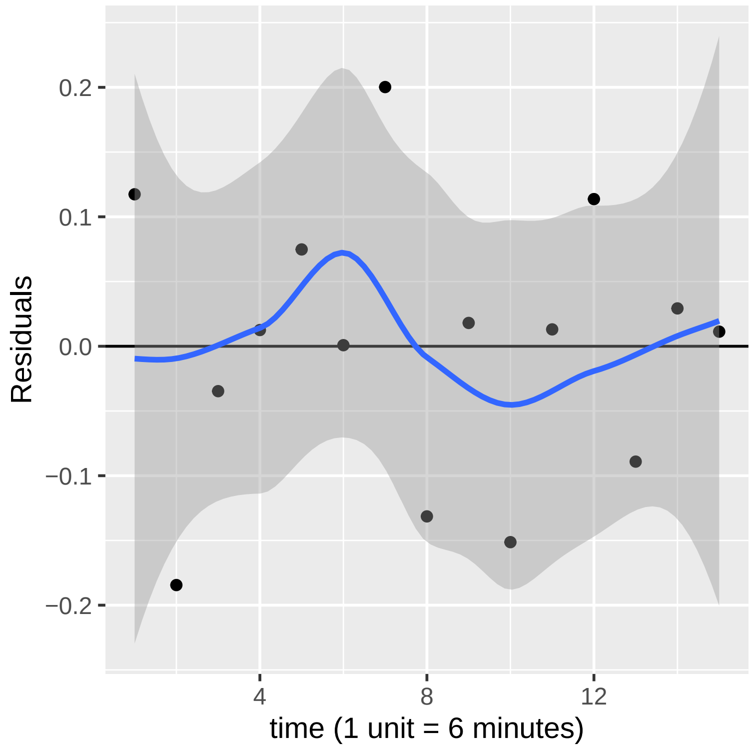
$$\log n_t = \log n_0 + \beta_1 t = \beta_0 + \beta_1 t$$

Which suggest we regress $\log n_t$ against t

```
lm2 = lm(log(N_t) ~ t, data=bact)
ggplot(bact, aes(x=t, y=log(N_t)))+geom_point()+
geom_smooth(method='lm', se=F)+xlab("time (1 unit = 6 minutes)")+
ylab("log of count of surviving\nbacteria (in 100s)")
```




```
ggplot(bact, aes(x=t, y=lm2$res))+geom_point() +
xlab("time (1 unit = 6 minutes)")+ ylab("Residuals")+
geom_hline(yintercept=0) + geom_smooth()
```



- Scatter plot shows transformation achieves linearity
- Residual plot shows no clear violation of model assumptions

3.3 Transformation to Reduce Skewness

3.3.1 Why worry about skewness

- If the response is skewed, the normality assumption of the noise ϵ is probably violated.
- If a predictor is highly-skewed, there might be extreme outliers or influential points. Transforming the predictor might make the extreme outliers less extreme and reduce the impact of influential points.

Skewness can often be ameliorated by a power transformation.

$f_\lambda(y) = y^\lambda \ \ if \ \ \lambda \neq 0 \ \ and \ \ \log(y) \ \ if \ \ \lambda = 0$

- If **right-skewed** try taking square root, logarithm or other powers
- If **left-skewed** try squaring, cubing

The square-root transformation can shorten the upper tail and extend the lower tail, of a distribution and hence can reduce right-skewness.

Logarithm can shorten the upper tail and extend the lower tail even more.

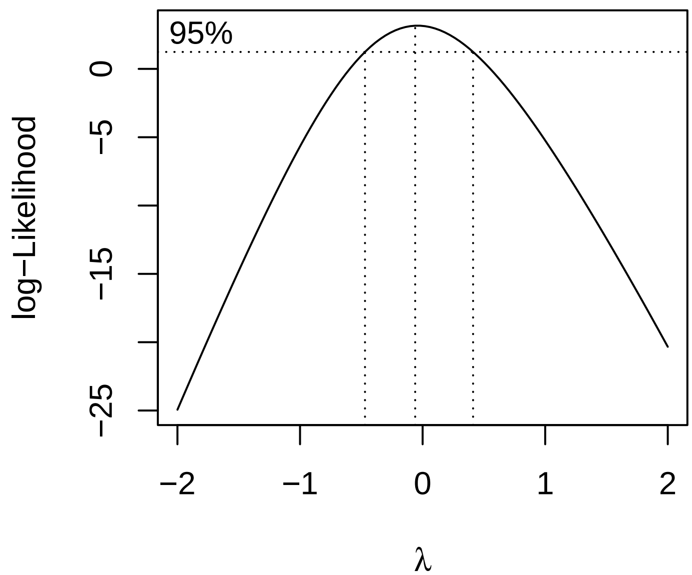
3.4 Box Cox Method

Box-Cox method is an automatic procedure to select the best power λ that make the residuals of the model

$$Y^\lambda = \beta_0 + \beta_1 X_1 + ... + \beta_p X_p + \epsilon$$

closest to normal and constant variability.

```r
library(MASS)
boxcox(lm(Y ~ X + I(X^2), data=supvis))
```



We use the log transformation since we choose $\lambda = 0$

Transformations are useful tools – we transform (rescale, generally) the variables in the model so that the linear regression model becomes (more) appropriate.

Transformations, however, cannot fix all problems:

- a non-linear model may be needed,
- one may try using weighted least square

3.4.1 Caution on transforming Variables

- Transformed variables might be difficult to interpret
- Remember – whenever you transform your variables, all your estimates and confidence intervals are expressed in that scale. To report your results, you need to convert BACK to the original scale.